

A randomized, placebo-controlled trial of acetaminophen for treatment of migraine headache.



OBJECTIVE: To evaluate the efficacy and safety of acetaminophen 1000 mg for the treatment of episodic migraine headache. **BACKGROUND:** While acetaminophen is commonly used to treat migraine, there have been limited published clinical trial efficacy results. **DESIGN/METHODS:** Ten investigators at 13 private, ambulatory, primary care sites in the United States enrolled and treated 346 outpatient adults 18-72 years of age with migraine headache of moderate to severe intensity into a randomized, placebo-controlled, double-blind clinical trial of 6 hours duration. Each patient was randomly assigned to a single dose of study medication of acetaminophen 1000 mg ($n = 177$) or placebo ($n = 169$). The percentage of patients with a reduction in baseline headache pain intensity from severe or moderate to mild or none 2 hours after treatment and the headache pain intensity difference from baseline at 2 hours were the primary efficacy measures. Other measures of pain relief, severity differences from baseline for migraine-associated symptoms of nausea, photophobia, phonophobia, and functional disability, and percentage of patients with migraine-associated symptoms reduced to none were also assessed. **RESULTS:** Significantly ($P = .001$) more patients treated with acetaminophen 1000 mg reported mild to no pain after 2 hours (52.0%) compared with those treated with placebo (32.0%). The mean pain intensity difference from baseline measured at 2 hours was significantly ($P < .001$) greater for patients treated with acetaminophen 1000 mg (0.82) compared with those treated with placebo (0.46). A significant difference in favor of acetaminophen 1000 mg over placebo was also observed at 1 hour after treatment for the percentage of patients with mild to no pain and for mean pain intensity difference from baseline. Acetaminophen 1000 mg was significantly more effective than placebo for all but 1 (pain reduced to none at 2 hours) clinically important secondary pain relief outcomes. Mean severity changes from baseline in migraine-associated symptoms of nausea, photophobia, phonophobia, and functional disability at 2 and 6 hours were significantly ($P < .001$) in favor of acetaminophen over placebo; the percentage of patients with no symptoms at 2 and 6 hours statistically significantly favored acetaminophen in 6 of 8 comparisons. Adverse events, overall, and specifically for nausea, were reported more frequently in the placebo group. **CONCLUSIONS:** Acetaminophen 1000 mg, a nonprescription drug, is an effective and well-tolerated treatment for episodic and moderate migraine headache. In addition, acetaminophen generally provided a beneficial effect on associated symptoms of migraine including nausea, photophobia, phonophobia, and functional disability.